

UTKARSH SINGH

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Professional Summary

Second-year Computer Science Engineering student specializing in AI, driven by the craft of turning ambitious ideas into working systems. I build practical AI tools across agentic workflows, backend infrastructure, and applied machine learning.

Education

Maharaja Agrasen Institute of Technology [GGSIPU] New Delhi, India
B.Tech in Computer Science Engineering, Specialization in Artificial Intelligence Aug. 2025 – May 2029
(9.3 CGPA)

Kamal Public Sr. Sec. School [CBSE] New Delhi, India
Class XII PCM with Computer Science (92.4%) April 2011 – May 2025

Skills

Programming Languages: Python, TypeScript, JavaScript, SQL, C++, C

AI & Machine Learning: Generative AI, Agentic AI, Multi-Agent Systems, RAG, Multimodal RAG, LLM Applications, Deep Learning, PyTorch, TensorFlow, Transformers, Hugging Face, scikit-learn

Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs **Frontend:** React, Next.js, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Redis, Firebase, SQLite

MLOps & Cloud: Docker, Git, GitHub Actions (CI/CD), AWS (S3, EC2, Lambda), GCP, Vercel, Render, Railway

Libraries & Tools: NumPy, Pandas, LangChain, Matplotlib, OpenCV, Postman, Jupyter

Projects

KAVACH — Fraud Risk Manager for Indian D2C Merchants

- Built a defence-only risk manager pairing a calibrated XGBoost return-risk scorer with NetworkX graph detection over shared phones, devices, VPAs and fuzzy-matched addresses to catch COD return abuse and coordinated fraud rings.
- Ring detection scored 0.889 precision at 1.000 recall on held-out data; every decision ships with SHAP attributions and a SHA-256 hash-chained audit trail.

NETRA — AI Growth Copilot for Indian Kirana Stores

- AI copilot delivering demand forecasting, non-collusive price benchmarking and credit scoring to merchants over Soundbox audio, WhatsApp and web in 7 Indic languages.
- Enforces a five-layer privacy boundary: k-anonymity cohort gating, Laplace differential privacy, and an anti-reconstruction interceptor that blocks competitor surveillance and price collusion.

ATLASS — Adaptive Taste Learning And Suggestion System

- Hybrid movie recommender running entirely in-browser with zero inference server; SVD fold-in blended 70/30 with a 48-dimensional LSA content profile, pre-trained weights shipped as ~700KB JSON.
- Built on MovieLens (9,742 films, 100K+ ratings) and fronted by a WebGL gallery rendering cards along a parametric arc.

SCOUT — AI-Powered Cross-Platform Shopping Assistant

- AI shopping agent that concurrently searches seven Indian e-commerce platforms (Amazon, Flipkart, Blinkit, Zepto, Instamart, Myntra, Ajio), deduplicates results by title fingerprint, and ranks on a six-factor weighted score.
- Async FastAPI backend with an LLM-based intent router handling Hindi/Hinglish with slang and typos, per-session Redis state, and SSE streaming for real-time recommendations.

Achievements

Top 10 Finalist — Paytm Build for India AI Hackathon 2026

- Finished Top 10 nationally; built NETRA, an AI growth copilot for Indian merchants with privacy-preserving market intelligence using k-anonymity and differential privacy.

Published Researcher at ICML (International Conference on Machine Learning) 2026

- Co-authored **Adaptive Termination in Agentic Multimodal RAG via Cross-Model Evidence Grounding**, accepted at ICML 2026 (~26% acceptance rate); published as a second-year undergraduate.

Honoured — Award of Excellence 2025

- Honored with the Academic Excellence Award, presented by Shri Suresh Jain (National Organization Secretary, Bharat Vikas Parishad) and Shri Vijay Kumar (Joint Commissioner of Police).

Finalist — ET Gen AI Hackathon 2026

- Co-built an Autonomous Multi-Agent AI Code Intelligence & Remediation Platform; ranked Top Performing out of 3000+ candidates nationwide.